

Homework 6

Solve the following problems justifying all your answers. You should not use a graph calculator to justify any answer.

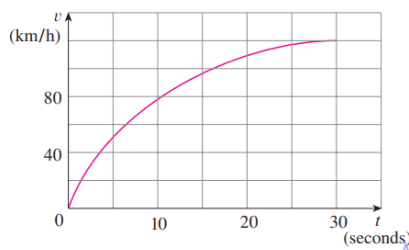
1. Find the most general antiderivative of the following functions:

(a) $f(x) = \frac{2x^4 + 4x^3 - x}{x^3}$, $x > 0$,
 (b) $g(x) = 2^x + 4 \sinh x - \frac{3}{\sqrt{1-x^2}}$.

2. A particle is moving with the given data. Find the position of the particle:

$$a(t) = 10 \sin t + 3 \cos t, \quad s(0) = 0, \quad s(2\pi) = 12$$

3. The velocity graph of a car accelerating from rest to a speed of 120 km/hr over a period of 30 second is shown. Estimate the distance traveled during this period.



4. Evaluate the following integral by Reimman sums

$$\int_{-2}^0 (x^2 + x) \, dx$$

5. Use the properties of integrals to verify the inequality without evaluating the integrals:

$$2 \leq \int_{-1}^1 \sqrt{1+x^2} \, dx \leq 2\sqrt{2}$$

6. If $f(x) = \int_0^{\sin x} \sqrt{1+t^2} \, dt$ and $g(y) = \int_3^y f(x) \, dx$, find $g''(\frac{\pi}{6})$.

7. Evaluate the following integrals:

(a) $\int_0^1 x (\sqrt[3]{x} + \sqrt[4]{x}) \, dx$,
 (b) $\int_0^{\frac{\pi}{4}} \frac{1+\cos^2 x}{\cos^2 x} \, dx$,
 (c) $\int_0^1 \frac{4}{1+x^2} \, dx$

8. Evaluate the following integrals:

(a) $\int \cos^3 x \sin x \, dx$,
 (b) $\int e^x \sqrt{1+e^x} \, dx$
 (c) $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} \, dx$,

(d) $\int_0^a \cos x \sin(\sin x) dx$

9. Sketch the region enclosed by the given curves in the first quadrant and compute the area of this region:

$$y = \frac{1}{x}, y = x, y = \frac{1}{4}x, x > 0$$